NTNU

Department of computer science

Examination paper for TDT4175 Information Systems

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Examination date: 04-12-2021

Examination time (from-to): 9:00-12:00

Permitted examination support material: *Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book* "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been specified. Clean copies of the chapters 1-6 used in the course also allowed. *In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.*

Language: English

Number of pages (front page excluded): 4

Number of pages enclosed: 5

Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig X **2-sidig □**

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skal ha flervalgskjema □

Checked by:

23/11/2021 Patrick Mikalef

Date Signature

Task 1 - (20 points)

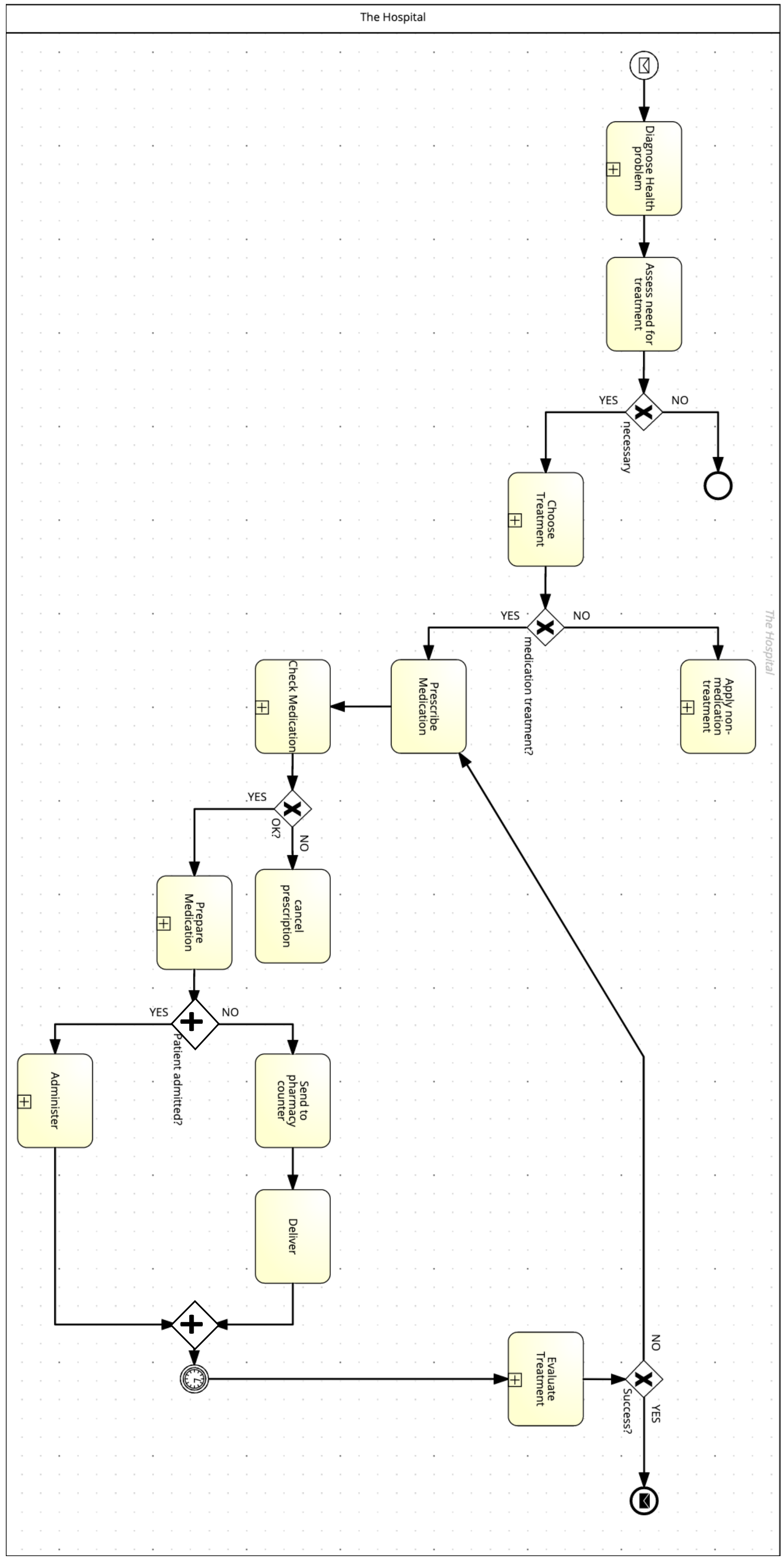
The most common type of therapy in health care is to prescribe medication to patients. Medication can be provided to patients in many different ways. It can come in the form of pills that have to be swallowed, fluid that the patient must drink, fluid that is injected, and so on. Because medication treatment is so widely spread it is also a main source for errors. Patients can get the wrong medication, a dose that is too high or medication that can't be used together with other medication they already use.

Because of its importance in the delivery of care, a big hospital decides to support the process of prescribing and giving medication (also known as 'medication management') by means of an information system. As you have learned, the first step in developing such a system is to get a thorough understanding of the medication management process as it is.

Based on a diagnosis of the patient's health problem, a doctor prescribes medication. A nurse is not allowed to do this. The prescription is written on a special form that is sent to the hospital's pharmacy. Here the prescription is checked by a pharmacist. If the prescribed medication does not fit other health problems the patient has or does not go together with medication already used, the prescription is cancelled, and the doctor is asked to revise it. If it is ok, the pharmacy prepares the medication. If the patient is admitted to the hospital, it is sent to the department where the patient is hospitalized. If the patient is not in the hospital it is sent to the pharmacy's counter where it can be collected by the patient.

The medication is given to or taken by the patient according to the prescription. This is called administering the medication. For example, the patient swallows a pill three times a day for a week. Or in the hospital a fluid is injected by a nurse every day for five days in a row. After the medication has been taken (note that this can take several days or even weeks), the patient's health status is checked to see if the treatment works. This is done by the doctor. If the treatment seems to work but the health problem has not disappeared completely, the treatment will be prolonged. This basically means that a new prescription is issued. If the health problem has disappeared, the treatment is finished. If it proves that the treatment does not work, it is also finished. The health problem of the patient must be assessed again because the diagnosis was apparently wrong.

The model given below contains several mistakes, both syntactical and semantical mistakes. Find at least three syntactical and three semantical mistakes (so, six mistakes in total). Motivate why you consider these to be mistakes.

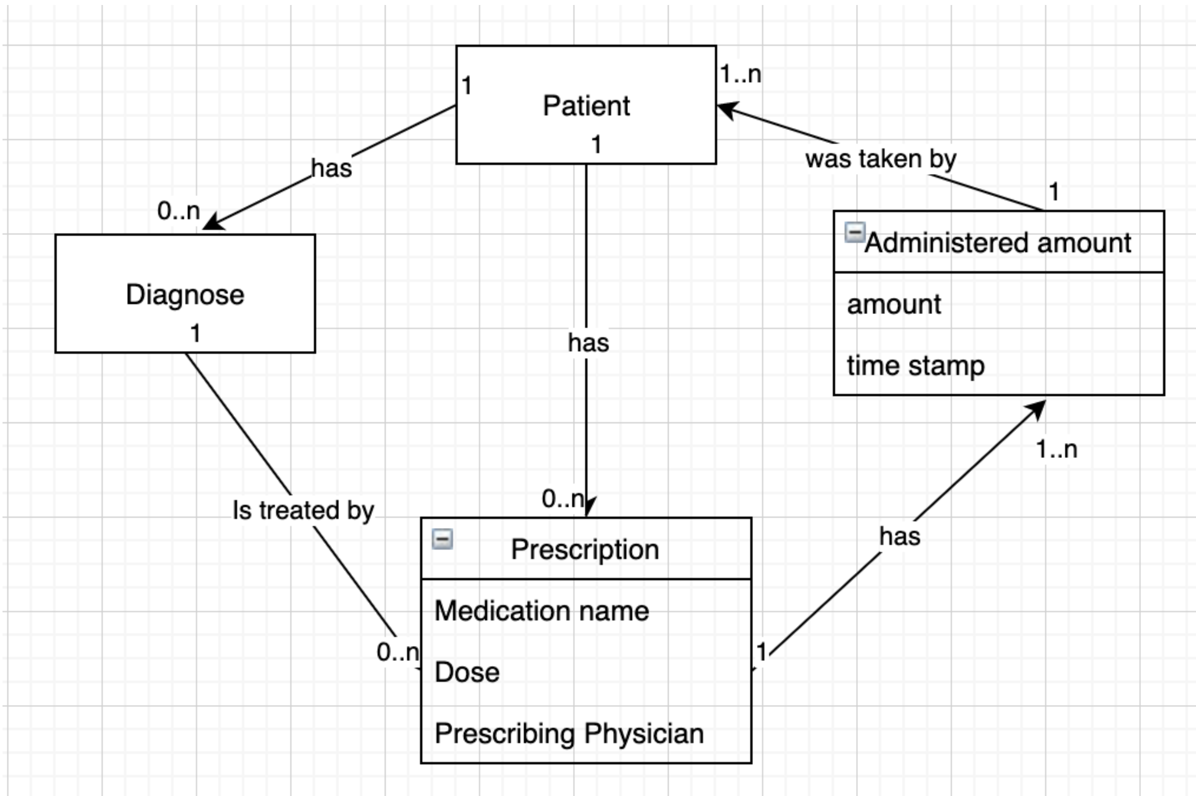


The Hospital
Diagnose Health problem
Assess need for treatment
YES
NO
necessary
Choose Treatment
YES
NO
medication treatment?
Apply non-medication treatment
Prescribe Medication
Check Medication
.YES
NO
OK?
cancel prescription
Prepare Medication
YES
NO
Patient admitted?
Administer
Send to pharmacy counter
Deliver
Evaluate Treatment
NO
YES
Success?
The Hospital

Task 2 (5 points)

What is a computer-based information system (CBIS)?

Task 3 (15 points)



The simple data model above represents the information that will be at the core of the medication management system that will be developed and implemented. In the first lecture and in chapter 1 of the book, quality of information was discussed. A table was presented with 11 characteristics of information quality. Discuss at least three of these characteristics that you consider to be crucial for the success of the medication management system.

Task 4 (5 points)

What is a decision support system (DSS)?

Task 5 (15 points)

One of the main goals with the development and implementation of the medication management system is to increase patient safety. That includes that a patient gets the right diagnose as soon as possible, the right treatment and that mistakes are reduced to a minimum. One way to achieve this is to add ***decision support system (DSS)*** functionality to the system. Which parts of the process presented in Task 1 would you support by DSS functionality? Describe briefly how this functionality supports the decision making. Note that DSS functionality could be useful at several places in the process.

Task 6 (10 points)

How would you characterize the introduction of the medication management system: as an example of ***reengineering*** or ***continuous improvement***? Motivate your answer.

Task 7 (10 points)

As part of the hospital's quality management program information on unwanted deviations or mistakes is collected from the medication management system. Examples of this information are:

1. Patients being diagnosed wrongly
2. The wrong treatment given a diagnose
3. Medication was prescribed that could not be combined with medication the patient already had
4. The wrong medication given to a patient
5. The wrong amount of medication given to a patient

The collected information is used in an information system (system X) that can aggregate, combine and visualize it. The book and the lectures use a typology of information systems that includes: Transaction Processing Systems; Customer Relationship Management Systems; Supply Chain Management Systems; Decision Support Systems; Management Information Systems; Executive Support Systems. How would you classify system X? Motivate your answer.